

Trimoyee Ghosh

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Summary

Backend-focused Software Engineer with hands-on experience building distributed systems, event-driven microservices, and RAG/LLM-integrated applications. Recent B.E. graduate and immediate joiner for SDE-1 roles.

Experience

Louisa AI

Software Engineer Intern

Jan 2026 – Present

Bengaluru, Karnataka, India

- Engineered a **browser extension** for news platforms using **React**, **TypeScript (ES6+)** and **Plasmo**, enabling seamless in-page AI interaction via a draggable floating responsive UI with cross-tab state sync; worked in Agile sprints.
- Built a robust messaging layer across background, content, and UI scripts with backend API integration.
- Implemented authentication and auth-based conditional rendering for secure and dynamic user experience.

Scale AI

Freelance AI Quality Evaluator

May 2026 – Present

Remote

- Evaluated pre-release **AI models** via **RLHF** feedback across reasoning, factual accuracy, and logical consistency.
- Designed adversarial prompts to surface hallucination and instruction-following gaps across multimodal tasks.

PwC

Technology Consultant Intern

May 2025 – July 2025

Kolkata, West Bengal, India

- Analyzed end-to-end **ETL** workflows for data migration pipelines, documenting data flow across different stages.
- Developed **Power BI** dashboards with DAX queries, transforming raw datasets into KPI reports.

Projects

SafeHer | *React, TypeScript, Java, Spring Boot, Python, FastAPI, PostgreSQL, MongoDB, Kafka, LangChain, Docker* | [GitHub](#)

- Architected an **8-service microservices platform** for crowdsourced women's safety ratings with **JWT** auth, **Elasticsearch** fuzzy search, and **Kafka**-decoupled services; containerized with **Docker Compose**.
- Built Python/FastAPI service for integrating **LLM agents (LangChain)** for automated review moderation, fake-rating anomaly detection (**Redis** velocity gating), and safety summarization.
- Implemented a dual-mode **RAG** chatbot with a **LangGraph** self-grading loop: **vectorless** live-fetch for place-specific queries; **pgvector semantic search** for open discovery queries; async batch reviews embedding via **Kafka**.

URL Shortener | *React, JavaScript, Java, Spring Boot, RabbitMQ, Redis, MySQL, Docker, JUnit 5* | [GitHub](#)

- Developed a **high-throughput and scalable URL shortener** with **Snowflake IDs**, **Base62 encoding**, async click analytics via **RabbitMQ**; **Redis** caching with **Cuckoo Filter**, and **Lua-scripted** leaky bucket rate limiting; **JWT** auth.
- Achieved **1232 req/sec** peak and **sub-139ms P95 latency**; validated via **k6** load testing and **Prometheus/Grafana**.
- Tested with **JUnit 5**, **Mockito** (unit, integration, E2E); automated **CI** via **GitHub Actions** with **Docker Hub** push.

Deton8 | *Next.js, Node.js, Express.js, Socket.io, TypeScript, Docker, CI/CD* | [GitHub](#) | [Live Demo](#)

- Built a **full-stack multiplayer board game** with real-time **WebSocket** communication, room-based matchmaking, AI opponents (greedy heuristic), and backend-authoritative move validation.
- Automated **CI/CD** with **GitHub Actions**, **Docker** and **Docker Hub**; **cloud-deployed** on Railway.

Achievements

- National Semi-Finalist in **Flipkart Grid 7.0** among 160,000+ participants.
- LeetCode Knight** (max rating: **1957**, Top **3.24%** globally); solved 1,200+ coding problems. [LeetCode Profile](#)
- Achieved **Global Rank 436** in **LeetCode Biweekly Contest 185** among 39,200+ participants.
- Secured **Global Rank 25** in **CodeChef Starters 244 Div 2** among 1,100+ participants. [CodeChef Profile](#)

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Node.js, Express.js, Spring Boot, Tailwind CSS, FastAPI, Socket.io, LangChain, JUnit 5

Concepts: REST APIs, Microservices, Distributed Systems, Object-Oriented Programming, Event-Driven Architecture, RAG, CI/CD

Databases & Tools: MongoDB, MySQL, Git, GitHub, GitHub Actions, Redis, Docker, Kafka, RabbitMQ, Elasticsearch

Education

Jadavpur University

Bachelor of Engineering in Mechanical Engineering

Kolkata, West Bengal, India

Nov 2022 – June 2026

G.D. Birla Centre for Education

Higher Secondary (ISC)

Kolkata, West Bengal, India

May 2022